

ESE 101 Homework 3

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1 Greenhouse Effect

Consider an atmosphere consisting of n homogeneous layers in radiative equilibrium with an underlying surface. Assume the atmosphere absorbs no solar radiation, the surface has albedo α , and the atmospheric layers absorb and emit as blackbodies (emissivities $\epsilon = 1$).

Part 1

What is the radiative energy balance at the top of the atmosphere and in each atmospheric layer?

For the system to be in radiative equilibrium, the energy flux in must equal the energy flux out for each component (**Taylor section 6.8**). The problem states layers are blackbodies, so they absorb all longwave radiation that hits them and emit σT^4 upward and downward (**Taylor section 6.8**). The atmosphere is transparent to solar radiation, so all incoming solar energy $(1 - \alpha)S_0/4$ reaches the surface. Let T_i be the temperature of layer i (with $i = 1$ being the top layer) and T_s be the surface temperature.

- **Top of Atmosphere (TOA):** The incoming solar radiation absorbed by the planet $(1 - \alpha)S_0/4$ must be balanced by the longwave radiation escaping to space, which can only come from the top layer (Layer 1).
- **Layer i :** Any layer i (for $1 \leq i < n$) receives longwave radiation from the layer above it ($i - 1$, or space=0 for Layer 1) and the layer below it ($i + 1$). It emits as a blackbody both up and down ($2\sigma T_i^4$).
- **Layer n (Bottom):** The bottom layer receives longwave from Layer $n - 1$ above and from the surface T_s below. It also emits up and down.
- **Surface:** The surface receives the full solar radiation $(1 - \alpha)S_0/4$ and longwave radiation from the bottom layer n . It emits its own longwave radiation upward.

This gives the following balance equations:

$$\begin{aligned}\text{TOA: } (1 - \alpha)\frac{S_0}{4} &= \sigma T_1^4 \\ \text{Layer 1: } \sigma T_2^4 &= 2\sigma T_1^4 \\ \text{Layer } i \ (1 < i < n): \quad \sigma T_{i-1}^4 + \sigma T_{i+1}^4 &= 2\sigma T_i^4 \\ \text{Layer } n: \quad \sigma T_{n-1}^4 + \sigma T_s^4 &= 2\sigma T_n^4 \\ \text{Surface: } (1 - \alpha)\frac{S_0}{4} + \sigma T_n^4 &= \sigma T_s^4\end{aligned}$$

Part 2

What is the radiative equilibrium temperature for each layer and the surface as a function of n ? What are representative values for Earth in the top layer of the atmosphere and at the surface for a few values of n (total solar irradiance $S_0 = 1362 \text{ W m}^{-2}$ and albedo $\alpha = 0.3$)?

We can solve the system of $n + 1$ balance equations from **part 1** by inspection and substitution. Let $F_{in} = (1 - \alpha)S_0/4$.

- From the TOA equation: $\sigma T_1^4 = F_{in}$.
- From the Layer 1 equation: $\sigma T_2^4 = 2\sigma T_1^4 = 2F_{in}$.
- The general equation for Layer i ($\sigma T_{i-1}^4 + \sigma T_{i+1}^4 = 2\sigma T_i^4$) shows that the emitted flux increases linearly for each layer as we move down.
- By induction, this gives a general solution for any layer i : $\sigma T_i^4 = i \cdot F_{in}$.
- We use this general solution in the surface balance equation: $F_{in} + \sigma T_n^4 = \sigma T_s^4$.
- Substituting T_n : $F_{in} + (n \cdot F_{in}) = \sigma T_s^4$, which simplifies to $\sigma T_s^4 = (n + 1)F_{in}$.

This multi-layer model is a form of a “grey” atmosphere model, where we assume the atmosphere acts as a blackbody at long wavelengths (**Taylor section 6.8**). The general temperature functions are:

$$T_i(n) = \left[\frac{i(1-\alpha)S_0}{4\sigma} \right]^{1/4} \quad T_s(n) = \left[\frac{(n+1)(1-\alpha)S_0}{4\sigma} \right]^{1/4}$$

For the representative values, we first calculate the absorbed solar flux: $F_{in} = (1 - 0.3) \times (1362 \text{ W m}^{-2})/4 \approx 238.35 \text{ W m}^{-2}$. The Stefan-Boltzmann constant is $\sigma \approx 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

The top layer temperature T_1 is constant for any $n \geq 1$, as it must balance the incoming solar radiation:

$$T_1 = \left[\frac{1 \times 238.35 \text{ W m}^{-2}}{5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}} \right]^{1/4} \approx 254.6 \text{ K}$$

The surface temperature T_s increases with n , the number of atmospheric layers:

Layers (n)	Surface Temperature $T_s(n)$
1	$T_s(1) = (1 + 1)^{1/4} \times T_1 \approx (1.189) \times 254.6 \text{ K} \approx 302.7 \text{ K}$
2	$T_s(2) = (2 + 1)^{1/4} \times T_1 \approx (1.316) \times 254.6 \text{ K} \approx 335.0 \text{ K}$
5	$T_s(5) = (5 + 1)^{1/4} \times T_1 \approx (1.565) \times 254.6 \text{ K} \approx 398.5 \text{ K}$

Part 3

Provide a (narrative) physical explanation for the dependence (or lack thereof) of the temperature of the top layer and of the surface on n .

Top Layer Temperature (T_1)

The temperature of the top layer, T_1 , is **independent of n** because it is solely determined by the top-of-atmosphere (TOA) energy balance. In this model, the atmosphere is transparent to solar radiation, but opaque (a blackbody) to longwave radiation. This means the only way the planet can lose energy to space is by the upward emission from the single top layer. For the entire planet to be in equilibrium, this outgoing longwave flux (σT_1^4) must exactly balance the total incoming solar flux absorbed by the surface ($(1 - \alpha)S_0/4$). Since this balance does not involve any other layers, T_1 is fixed, regardless of how many layers n are stacked beneath it.

Surface Temperature (T_s)

The surface temperature, T_s , is strongly dependent on n . The surface is heated by two sources: the absorbed solar radiation (F_{in}) and the downward longwave radiation from the bottom atmospheric layer (σT_n^4). The atmosphere traps longwave radiation, creating a greenhouse effect (**Taylor section 6.7.3**). Each layer i must be warmer than the layer above it ($T_i > T_{i-1}$) to transfer energy upward and maintain equilibrium. As we add more layers (n increases), this insulating “blanket” becomes thicker. This requires the lowest layer n to be progressively warmer to drive the heat flux up through the stack. This much warmer bottom layer T_n radiates more energy back down to the surface. The surface must then become even hotter to balance both the solar energy and this increased downward longwave radiation. Therefore, T_s increases with n .

Part 4

The atmospheric layer adjacent to the surface may be physically thin. Yet there is a difference between the temperature of the surface and the temperature of this near-surface layer of the atmosphere. What causes it? What are some of its implications?

The temperature difference is caused by non-radiative energy fluxes that transfer heat between the surface and the atmosphere. Our blackbody model only considered radiative transfer, but in reality, the surface and the air layer directly exchange energy via:

1. **Sensible Heat Flux:** The surface, heated by the sun, transfers thermal energy to the air in direct contact with it via conduction. This warmer, less dense air then rises, carrying the heat upward via convection.
2. **Latent Heat Flux:** Energy is used at the surface to evaporate water (or transpire from plants). This removes heat, cooling the surface. The energy is stored as latent heat in the water vapor, which is then transported into the atmosphere by convection and released when the vapor condenses into clouds.

Implications

- **Convection & Weather:** This temperature difference is the primary driver of convection and turbulence in the lower atmosphere (the planetary boundary layer). This process mixes heat and moisture, forms clouds, and creates weather.
- **Lapse Rate:** A pure radiative equilibrium model, like the one in this problem, is often unstable near the surface (temperature drops too fast with height). Convection kicks in to mix the air, establishing a more stable temperature profile, known as the adiabatic lapse rate (**Taylor section 6.8**). This is a key reason our n -layer model predicts unrealistically high surface temperatures; it’s missing this convective “escape valve” for energy.
- **Hydrological Cycle:** The latent heat flux is a crucial part of the surface energy budget and is the primary mechanism that moves water from the surface into the atmosphere, driving the entire hydrological cycle.

2 Pressure Broadening

One of the ways radiation can interact with matter is by ABSORPTION. For a given adsorption line, the wavelengths/frequencies of light that can cause a molecule to undergo specific energetic transitions (e.g. electronic or vibrational) follow a distribution (in this context it is usually called a SPECTRAL LINE SHAPE). The width of one of these line shapes is determined by the finite lifetime of the excited state of a molecule. In this question, we will be thinking about how changes in pressure change these line shapes. From class, recall the Lorentz Line Profile:

$$\Phi_L(\nu, \alpha) = \frac{1}{\pi} \left[\frac{\alpha}{(\nu_0 - \nu)^2 + \alpha^2} \right]$$

Here, ν_0 is the peak frequency for the line shape and α is the HALF-WIDTH (the distance away from ν_0 when half of the peak height is achieved). We use the Lorentz line shape because it is the appropriate shape to describe spontaneous decay. We will save you from all of the math (see Canvas for PDF), but using the ideal gas law, Boltzmann's distribution, and Plank's Law, we can write the following expression for line half-width:

$$\alpha = \alpha_0 \left(\frac{p}{p_0} \right) \sqrt{\frac{T_0}{T}}$$

If we ASSUME the troposphere is adiabatic (this assumption gives us a way to link temperature and pressure in the atmosphere), we can further assume $T_0/T = (p/p_0)^{\frac{1-\gamma}{2\gamma}}$ (where $\gamma = c_p/c_v$ is the heat capacity ratio) and make a further substitution and get:

$$\alpha = \alpha_0 \left(\frac{p}{p_0} \right)^{\frac{1+\gamma}{2\gamma}} \quad \text{or} \quad \alpha = \alpha_0 \left(\frac{T}{T_0} \right)^{\left[\frac{\gamma+1}{2(\gamma-1)} \right]}$$

This equations tells us that if we know/measure (T_0, p_0, α_0) then I can calculate a new α for a given (T, p) .

Part 1

Calculate the ratio α/α_0 of half-widths for water at the surface (high pressure) $T_0 = 20^\circ\text{C}$ and water higher in the atmosphere (low pressure) $T = 1^\circ\text{C}$ (take $\gamma = 1.33$) and plot the Lorentz line shape for both temperatures as a function of multiples of α_0 . What does this tell you about the line widths for water near the surface v. high in the troposphere?

First, we calculate the ratio α/α_0 using the provided formula and values. We must convert the temperatures from Celsius to Kelvin:

- $T_0 = 20^\circ\text{C} = 293.15 \text{ K}$
- $T = 1^\circ\text{C} = 274.15 \text{ K}$

Now we can calculate the ratio of the half-widths:

$$\frac{\alpha}{\alpha_0} = \left(\frac{T}{T_0} \right)^{\left[\frac{\gamma+1}{2(\gamma-1)} \right]} = \left(\frac{274.15 \text{ K}}{293.15 \text{ K}} \right)^{\left[\frac{1.33+1}{2(1.33-1)} \right]} \approx 0.791$$

So, $\alpha/\alpha_0 = 0.791$. We'll use Python to generate the plot:

```
1 # Imports
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 def lorentz_normalized(x, alpha_ratio):
6     """
7     Calculates the normalized Lorentz line shape
```

```

8     x = (nu - nu_0) / alpha_0
9     alpha_ratio = alpha / alpha_0
10    """
11    numerator = alpha_ratio
12    denominator = x**2 + alpha_ratio**2
13    return numerator / denominator
14
15    # Set up the data
16    x = np.linspace(-4, 4, 400) # Normalized frequency axis
17    alpha_ratio_0 = 1.0          # Surface (T0 = 20C)
18    alpha_ratio_T = 0.791        # Atmosphere (T = 1C)
19
20    y_0 = lorentz_normalized(x, alpha_ratio_0)
21    y_T = lorentz_normalized(x, alpha_ratio_T)
22
23    # Create the plot
24    plt.figure(figsize=(8, 6))
25    plt.plot(x, y_0, color='red',
26            label=r'$T_0 = 20^\circ\text{C}$ (Surface, $\alpha = \alpha_0$)')
27    plt.plot(x, y_T, color='blue',
28            label=r'$T = 1^\circ\text{C}$ (Atmosphere, $\alpha = 0.791\alpha_0$)')
29
30    # Labels and title
31    plt.xlabel(r'Frequency difference from center $(\nu - \nu_0) / \alpha_0$')
32    plt.ylabel(r'Normalized Line Shape $(\Phi_L \cdot \pi \alpha_0)$')
33    plt.title('Lorentz Line Shape at Different Temperatures')
34    plt.legend()
35    plt.grid(True)

```

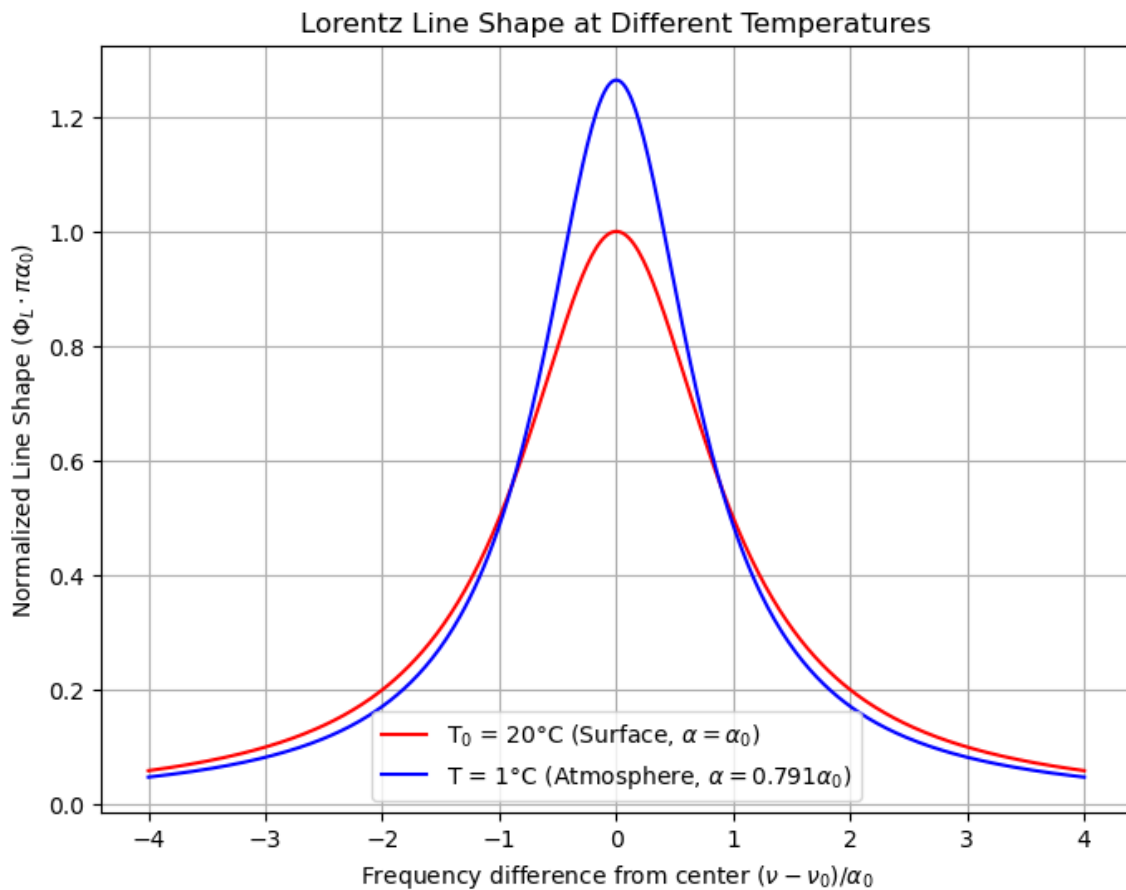


Figure 1: Lorentz profile for surface (T_0) and upper atmosphere (T) conditions.

This tells us that the absorption line for water is **broader** near the surface (high pressure/temp) and **narrower** higher in the troposphere (low pressure/temp). This phenomenon is known as **pressure broadening** (**Schneider section 4.3.3**). Because the pressure (and thus the rate of molecular collisions) is much higher near the

surface, the energy states of the molecule are perturbed, which “smears out” or broadens the absorption line (**Taylor section 6.7.3**). The plot also shows that the peak of the line is higher for the narrower case; this is because the total area under the curve (representing the total line strength) must be conserved.

Part 2

Explain two reasons why the UPPER atmosphere is generally transparent to radiation.

1. **Low Density of Absorbers:** The upper atmosphere is extremely sparse; the pressure and density of air molecules (including greenhouse gases like CO_2 and H_2O) are very low. Since absorption and scattering are physical interactions between photons and matter (**Schneider section 4.1**), the simple scarcity of molecules means there are far fewer opportunities for radiation to be absorbed or scattered. The “absorber amount” (**Taylor section 6.7.3**) in any given path is tiny, so most radiation passes through without interacting.
2. **Lack of Pressure Broadening:** Absorption lines, which are naturally very sharp, get “smeared” or broadened by molecular collisions. This pressure broadening is a dominant effect in the dense, high-pressure lower atmosphere, making the air opaque over wider frequency bands (**Schneider section 4.3.3**). In the low-pressure upper atmosphere, collisions are very rare. As a result, absorption lines remain extremely narrow. While the atmosphere is opaque at the precise center of these lines, the wide spectral “windows” between them are transparent, allowing most radiation to pass through unabsorbed (**Schneider section 4.3.3**).

3 More Radiation

Suppose today is September 22, 2023 (the Fall equinox) and you are in Pasadena, CA (34.15°N, 118.14°W).

Part 1

Using code from last week and the textbook to guide you, calculate and plot the solar zenith angle as a function of the time of day. You may assume solar noon occurs at 12pm (noon). Label all axes and put a title on the plot (**for all plots in the HW set**).

To calculate the solar zenith angle (θ), we use the formula from the textbook (**Schneider section 3.4.1, Eq. 3.18**):

$$\cos(\theta) = \sin(\phi) \sin(\delta) + \cos(\phi) \cos(\delta) \cos(\eta)$$

Where:

- ϕ is the latitude (34.15°N).
- δ is the declination angle. For the fall equinox, the declination angle is 0° (**Schneider section 3.4.1**).
- η is the hour angle, which measures the time of day relative to solar noon. Since the solar day is 24 hours (360°) and we assume solar noon is at 12:00 PM, the hour angle in degrees is $\eta = 15^\circ \times (\text{time} - 12)$ (**Schneider section 3.4.1**).

At the equinox ($\delta = 0$), the formula simplifies to $\cos(\theta) = \cos(\phi) \cos(\eta)$. Sunrise and sunset occur when $\theta = 90^\circ$, which for this case is at 6:00 AM ($\eta = -90^\circ$) and 6:00 PM ($\eta = +90^\circ$). We will plot the zenith angle between these times in Python:

```
1 # Latitude for Pasadena, CA
2 phi_deg = 34.15
3 phi_rad = np.deg2rad(phi_deg)
4
5 # Declination angle for fall equinox is 0 (Schneider section 3.4.1)
6 delta_deg = 0.0
7 delta_rad = np.deg2rad(delta_deg)
```

```

8
9 # We plot from sunrise (6:00) to sunset (18:00)
10 time_of_day = np.linspace(6, 18, 300)
11
12 # Calculate hour angle: eta = 15 degrees * (t - 12 hours)
13 # (Schneider section 3.4.1)
14 hour_angle_deg = 15 * (time_of_day - 12)
15 hour_angle_rad = np.deg2rad(hour_angle_deg)
16
17 # Calculate zenith angle using Eq. 3.18 from Schneider section 3.4.1
18 # cos(theta) = sin(phi)sin(delta) + cos(phi)cos(delta)cos(eta)
19 cos_theta = (np.sin(phi_rad) * np.sin(delta_rad)) + \
20             (np.cos(phi_rad) * np.cos(delta_rad) * np.cos(hour_angle_rad))
21
22 # Convert to degrees
23 zenith_angle_deg = np.rad2deg(np.arccos(cos_theta))
24
25 # Plot results
26 plt.figure(figsize=(10, 6))
27 plt.plot(time_of_day, zenith_angle_deg, linewidth=2)
28 plt.xlabel("Time of Day (hours)")
29 plt.ylabel("Solar Zenith Angle  $\theta$  (degrees)")
30 plt.title("Solar Zenith Angle in Pasadena, CA on Fall Equinox (Sept 22)")
31 plt.xticks(np.arange(6, 19, 1)) # Ticks every hour
32 plt.axvline(12, color='red', linestyle='--')
33 plt.grid(True)

```

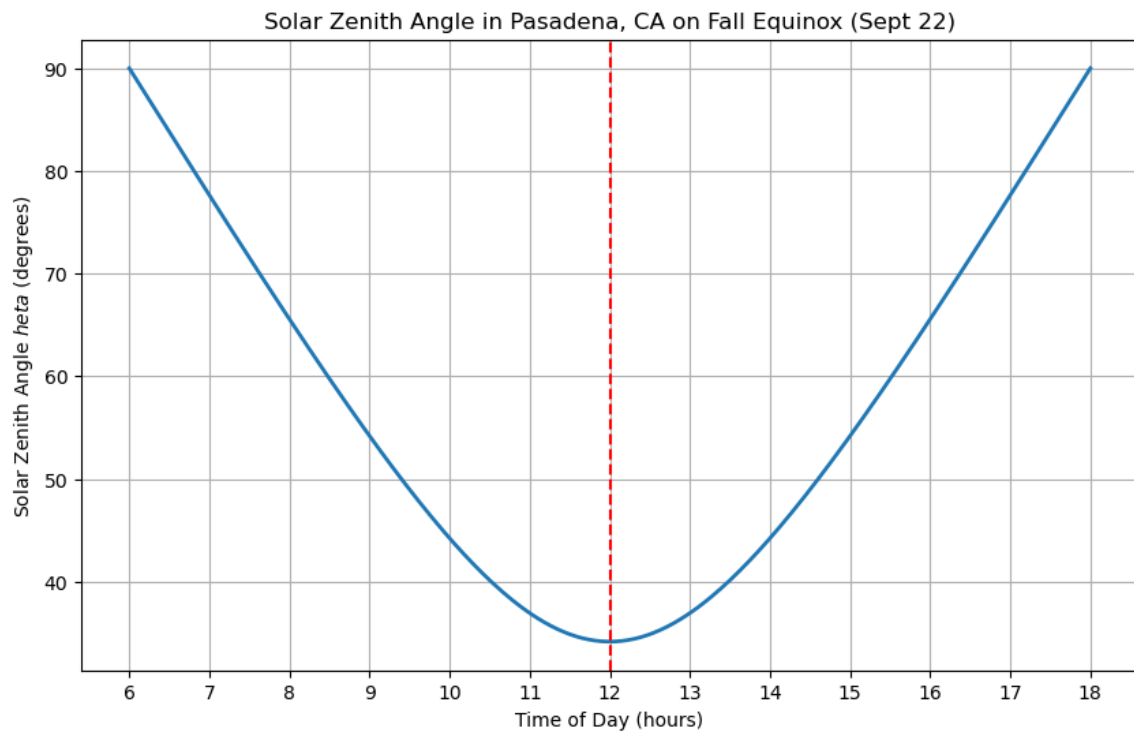


Figure 2: Solar zenith angle (θ) in Pasadena, CA (34.15°N) on the fall equinox. The minimum zenith angle at solar noon is 34.15° , which is equal to the latitude, as expected for an equinox.

Part 2

Now, make a plot of the top of atmosphere (TOA) insolation at this location as a function of time of data.

The top of atmosphere (TOA) insolation F is the solar radiative energy flux S that passes through a unit area tangential to the atmosphere. It is a function of the solar zenith angle θ (Schneider section 3.4, Eq. 3.15):

$$F = S \cos(\theta)$$

We are given the total solar irradiance $S_0 = 1362 \text{ W m}^{-2}$. While this value varies slightly during the year due to Earth's orbit (**Schneider section 3.3.1**), we will use $S \approx S_0$ as a constant. When the sun is set ($\theta > 90^\circ$), $\cos(\theta)$ is negative, and we set the insolation F to 0 (**Schneider section 3.4**).

We will build off of the code from **part 1** to calculate and plot this value:

```
1 S_0 = 1362 # Total Solar Irradiance (W/m^2)
2
3 # Calculate TOA insolation (Schneider section 3.4, Eq. 3.15)
4 # F = S * cos(theta)
5 toa_insolation = S_0 * cos_theta
6
7 # Plot results
8 plt.figure(figsize=(10, 6))
9 plt.plot(time_of_day, toa_insolation, linewidth=2, color='orange')
10 plt.xlabel("Time of Day (hours)")
11 plt.ylabel("TOA Insolation (W m$^{-2}$)")
12 plt.title("TOA Insolation in Pasadena, CA on Fall Equinox (Sept 22)")
13 plt.xticks(np.arange(6, 19, 1)) # Ticks every hour
14 plt.axvline(12, color='red', linestyle='--', label='Solar Noon (12:00 PM)')
15 plt.grid(True)
```

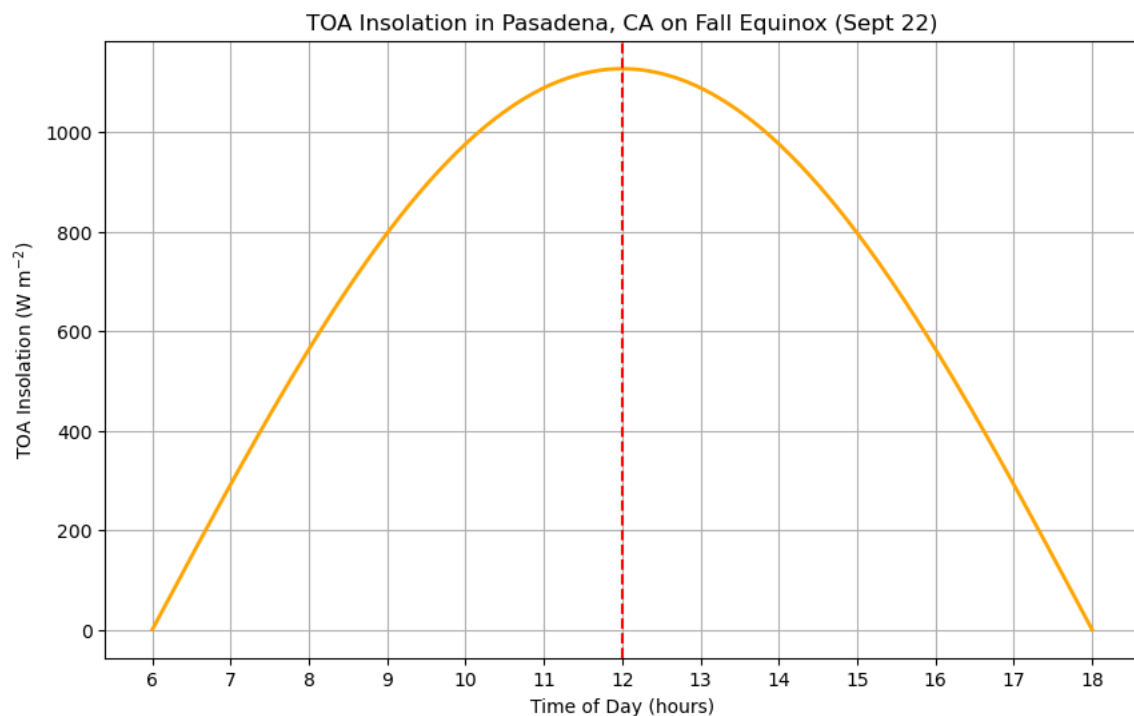


Figure 3: Modeled TOA insolation in Pasadena, CA on the fall equinox. The insolation is 0 at sunrise (6:00) and sunset (18:00) and peaks at solar noon (12:00).

Part 3

In the accompanying data file, you will find observed temperature and insolation recorded at Caltech (on the roof of Linde Laboratory to be specific). Make a plot of the observed insolation on the same plot as the TOA insolation you modeled above. What differences do you notice? Explain some possible reasons for this discrepancy. You may assume solar noon is the same as local noon since these weather reports use local time. In reality, solar noon was at 12:45pm local time on this day. You may account for this in your plots if you so choose.

We will now load the observed data from 2023-09-22-Temp.csv and plot it against the modeled TOA insolation from **part 2**. The Python code will first read the CSV file, convert the 'Time' column from 'HH:MM' format to decimal hours, and then plot both datasets:

```

1 # Imports
2 import pandas as pd
3
4 # Function to convert time to decimal hours
5 def time_to_decimal(time_str):
6     _, time = time_str.split()
7     hours, minutes = map(int, time.split(':'))
8     return hours + minutes / 60.0
9
10 # Load observed data
11 data = pd.read_csv('2023-09-22-Temp.csv')
12 # Convert 'Time' column
13 data['Decimal Time'] = data['Time'].apply(time_to_decimal)
14 observed_time = data['Decimal Time']
15 observed_insolation = data['Insolation']
16
17 # Use a consistent time array for the model
18 model_time = np.linspace(0, 24, 400) # Full 24-hour cycle
19 hour_angle_deg = 15 * (model_time - 12) # Model assumes noon at 12:00
20 hour_angle_rad = np.deg2rad(hour_angle_deg)
21
22 cos_theta = (np.sin(phi_rad) * np.sin(delta_rad)) + \
23             (np.cos(phi_rad) * np.cos(delta_rad) * np.cos(hour_angle_rad))
24
25 # When cos_theta < 0, sun is set. Clamp at 0.
26 cos_theta[cos_theta < 0] = 0.0
27
28 # Model TOA insolation
29 toa_insolation = S_0 * cos_theta
30
31 # Plot both model and observed data
32 plt.figure(figsize=(10, 6))
33 plt.plot(model_time, toa_insolation, linewidth=2, color='orange',
34          label='Modeled TOA Insolation')
35 plt.plot(observed_time, observed_insolation, linewidth=2, color='blue',
36          linestyle='-', marker='.', markersize=4,
37          label='Observed Surface Insolation (Linde Lab)')
38
39 plt.xlabel("Time of Day (hours)")
40 plt.ylabel("Insolation (W m-2)")
41 plt.title("TOA vs. Observed Insolation in Pasadena (Sept 22)")
42 plt.xticks(np.arange(0, 25, 1)) # Ticks every hour
43 plt.legend()
44 plt.grid(True)

```

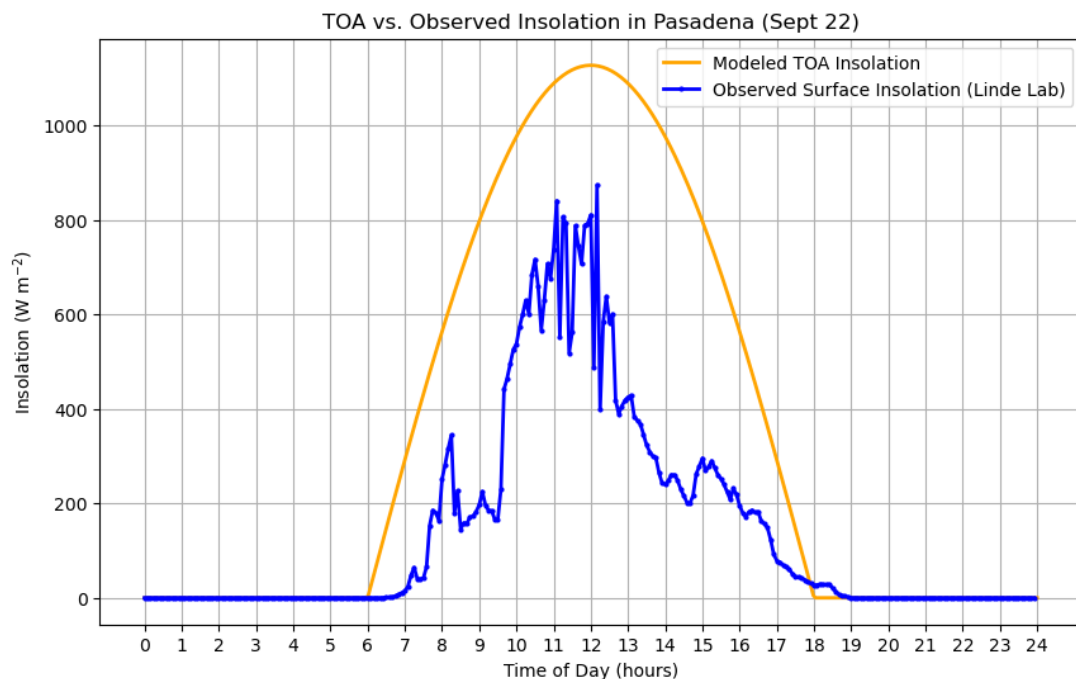


Figure 4: Modeled TOA insolation vs. observed surface insolation at Caltech on Sept 22, 2023.

There are two major differences between the modeled TOA insolation and the observed surface insolation:

1. **Magnitude:** The observed insolation at the surface is significantly lower than the TOA insolation at all times of day. The observed peak is around $800\text{--}900\text{ W m}^{-2}$, while the modeled TOA peak is over 1100 W m^{-2} .
2. **Shape:** The observed curve is not as smooth as the idealized TOA curve. It has small, high-frequency dips and jiggles.

The primary reason for the difference in magnitude is **atmospheric attenuation**. As sunlight passes through the atmosphere, a large fraction of it is either scattered or absorbed before it reaches the ground (**Schneider section 4.5**).

- **Scattering:** Air molecules (Rayleigh scattering) and aerosols/pollutants (Mie scattering) scatter photons in all directions, redirecting a portion of the solar beam away from the surface (**Schneider section 4.5.1, 4.5.2**). This scattering is why the sky is blue.
- **Absorption:** Atmospheric gases, primarily water vapor (H_2O), ozone (O_3), and CO_2 , absorb photons at specific wavelengths, converting their energy to heat (**Schneider section 4.1, 4.2**).

The irregular shape of the observed curve is most likely due to **clouds**. Even thin, high clouds (cirrus) or variations in atmospheric haze (aerosols) passing over the sensor will temporarily scatter or reflect more sunlight, causing the measured insolation to dip.

Part 4

Now, use the observed temperatures in the accompanying data file and the observed insolation on the Linde roof to make a plot of the estimated outgoing longwave radiation and compare it to the incoming shortwave radiation. Explain what you see in 2–3 sentences.

To estimate the outgoing longwave radiation (OLR), we use the Stefan-Boltzmann law, $F = \epsilon\sigma T^4$. We'll assume the surface (the Linde roof and surrounding area) acts as a blackbody, so its emissivity $\epsilon = 1$. We must use the temperature in Kelvin, so we convert the observed $T(C)$ to $T(K) = T(C) + 273.15$.

```
1 # Convert temperature to Kelvin
2 temp_kelvin = data['Temperature'] + 273.15
3 sigma = 5.67e-8 # Stefan-Boltzmann constant (W m^-2 K^-4)
4 epsilon = 1.0 # Assume emissivity (epsilon) = 1 (blackbody)
5
6 # Calculate OLR using Stefan-Boltzmann Law: OLR = epsilon * sigma * T^4
7 olr = epsilon * sigma * (temp_kelvin**4)
8
9 # Plot both datasets
10 plt.figure(figsize=(10, 6))
11
12 # Plot Incoming Shortwave (Observed Insolation)
13 plt.plot(data['Decimal Time'], observed_insolation, linewidth=2,
14          color='orange', label='Incoming Shortwave (Observed)')
15
16 # Plot Outgoing Longwave (Calculated)
17 plt.plot(data['Decimal Time'], olr, linewidth=2, color='blue',
18          label='Estimated Outgoing Longwave (OLR)')
19
20 plt.xlabel("Time of Day (hours)")
21 plt.ylabel("Radiative Flux (W m^-2)")
22 plt.title("Incoming Shortwave vs. Outgoing Longwave (Pasadena, Sept 22)")
23 plt.xticks(np.arange(0, 25, 1)) # Ticks every hour
24 plt.legend()
25 plt.grid(True)
```

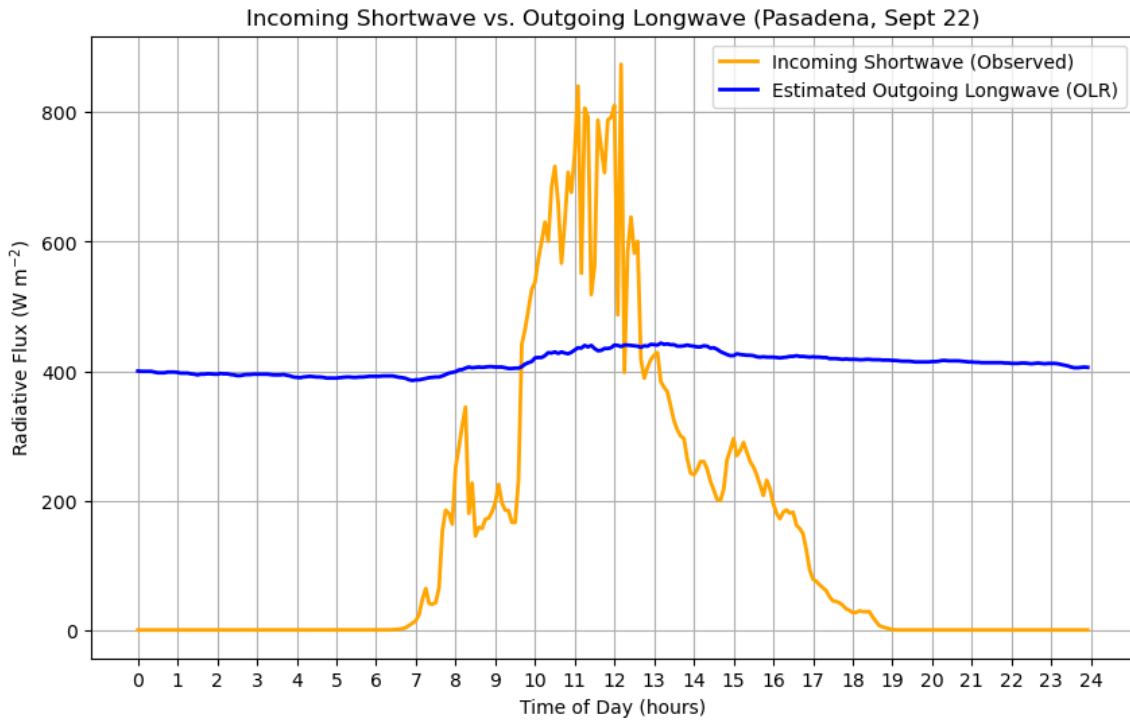


Figure 5: Observed incoming shortwave radiation (insolation) vs. estimated outgoing longwave radiation (OLR) calculated from observed temperature.

The plot shows that the incoming shortwave radiation (orange) follows a strong diurnal cycle, peaking at noon and being zero at night, as it is driven entirely by the sun. The outgoing longwave radiation (blue) is present 24 hours a day because all objects with a temperature above absolute zero emit radiation. The OLR is also much more stable (at around 400 W m^{-2}), following the smoother daily temperature curve: it peaks around noon/afternoon when the surface is warmest and reaches a minimum in the early morning right around sunrise.

Part 5

You might notice that the outgoing longwave radiation integrated over the day appears to be much more than the incoming shortwave radiation (feel free to take an integral to verify this!). But the temperatures appear relatively stable (i.e., temperatures aren't rapidly falling in the afternoon). Why might this be the case?

First, let's use Python to verify the integral (the total energy per square meter over the 24-hour period). We will use the trapezoidal rule for numerical integration.

```
1 # Imports
2 from scipy.integrate import trapezoid
3
4 # Convert time to seconds for Joules
5 time_seconds = data['Decimal Time'] * 3600.0
6
7 # trapezoid(y, x) computes the integral of y(x) dx
8 # Units are (W/m²) * (s) = J/m²
9 total_shortwave_energy = trapezoid(observed_insolation, time_seconds)
10 total_longwave_energy = trapezoid(olr, time_seconds)
11
12 # Convert to MegaJoules (MJ) for readability
13 total_sw_mj = total_shortwave_energy / 1e6
14 total_lw_mj = total_longwave_energy / 1e6
15
16 print(f"Total Incoming Shortwave Energy: {total_sw_mj:.2f} MJ/m²")
17 print(f"Total Outgoing Longwave Energy: {total_lw_mj:.2f} MJ/m²")
```

Total Incoming Shortwave Energy: 12.81 MJ/m²

Total Outgoing Longwave Energy: 35.45 MJ/m²

The integration confirms the observation: the surface is radiating almost three times as much energy (35.45 MJ/m^2) as it receives from the sun (12.81 MJ/m^2). The reason the temperature is stable is because the plot is missing the single largest energy source for the surface: downward longwave radiation (DLR), also known as “back radiation.”

The surface energy balance is not just (Incoming Shortwave) vs. (Outgoing Longwave). The full budget includes:

- Energy In = (Incoming Shortwave) + (DLR from atmosphere)
- Energy Out = (Outgoing Longwave Radiation) + (Sensible Heat Flux) + (Latent Heat Flux)

The atmosphere, filled with greenhouse gases like water vapor and CO_2 , absorbs the OLR from the surface (**Schneider section 4.2**). It then radiates this energy back in all directions, including downward. This DLR is a massive energy flux that is present 24/7, just like the OLR. On this day, the DLR was likely $\sim 20 \text{ MJ/m}^2$ (or $\sim 300\text{-}350 \text{ W/m}^2$ on average), making up the deficit.

Therefore, the temperature is stable because the total energy lost by the surface (OLR + sensible/latent heat) is balanced by the total energy gained (Insolation + DLR). The plot only compares one input to one output, revealing the power of the greenhouse effect.